

Problem Set 2

Data Visualisation for Social Scientists

Due: February 4, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Wednesday February 4, 2026. No late assignments will be accepted.

Study of Religious Congregations in Switzerland

The data for this problem set come from the National Congregations Study Switzerland (NCSS), which was conducted in 2008–2009 and 2022–2023. The data provide information on organisational structure, staffing, finances, worship practices, youth and educational activities, social composition, external engagement, and inclusion norms. The data were collected using stratified random samples of congregations drawn from comprehensive censuses, with interviews completed by a single knowledgeable key informant in each congregation, most often the spiritual leader.

Data Manipulation

1. Load the NCSS .csv file from GitHub into your global environment. Use the `select()` function to keep these variables in your dataframe:

```
1 NCSS <- read.csv("NCSS.csv")
2 head(NCSS)
3 library(dplyr)
4 NCSS_selected <- select(NCSS, CASEID, YEAR, GDREGION, NUMOFFMBR, TRAD6, TRAD12,
  INCOME)
```

I loaded the data into R as a csv file and isolated the key variables that are needed for the analysis.

- Congregation ID (CASEID)
 - Year (YEAR)
 - Region (GDREGION)
 - Number of official members (NUMOFFMBR)
 - 6-level religious classification (TRAD6)
 - 12-level religious classification (TRAD12)
 - Total income in last fiscal year (INCOME)
2. Filter the dataset so that you only include Christian, Jewish, and Muslim congregations (Chr étiennes, Juives, Musulmanes) using the TRAD6 variable.

```
1 Religion <- NCSS_selected %>%
2   filter (TRAD6 %in% c(" Juives" , " Musulmanes" , " Chr tiennes" ))
3 head ( Religion )
4 table ( Religion$TRAD6 )
```

Filtered the dataset so that only the three religions are shown in the TRAD6

	CASEID	YEAR	GDREGION	NUMOFFMBR	TRAD6
1	814	2022	Espace Mittelland	1300	Chr étiennes
2	5171	2022	R é gion lémanique	NA	Chr étiennes
3	3755	2022	Espace Mittelland	575	Chr étiennes
4	4636	2022	Zürich	2500	Chr étiennes
5	1331	2022	Espace Mittelland	NA	Chr étiennes
6	4267	2022	Espace Mittelland	2000	Chr étiennes

	TRAD12	INCOME
1	Protestantes réformées	NA
2	Catholiques romaines	NA
3	Catholiques romaines	NA
4	Catholiques romaines	NA
5	Évang éliques (conservatrices)	NA
6	Catholiques romaines	NA

> table (Religion\$TRAD6)

Chr étiennes	Juives	Musulmanes
1974	31	106

3. Compute for the number of congregations by religious classification (TRAD6) in each year, as well as the mean and median total income in last fiscal year (INCOME) by religious classification and year.

```

1 Religion_classification <- Religion %>%
2   filter(TRAD6 %in% c("Juives", "Musulmanes", "Chr étiennes")) %>%
3   group_by(YEAR, TRAD6) %>%
4   summarise(total_members=sum(NUMOFFMBR, na.rm=TRUE)) %>%
5   arrange(YEAR, desc(total_members))
6 Religion_classification
7 last_year <- max(Religion$YEAR, na.rm = TRUE)
8 Income_last_year <- Religion %>%
9   filter(TRAD6 %in% c("Juives", "Musulmanes", "Chr étiennes"),
10         YEAR == last_year) %>%
11   group_by(TRAD6) %>%
12   summarise(
13     mean_income = mean(INCOME, na.rm = TRUE),
14     median_income = median(INCOME, na.rm = TRUE),
15     count = sum(!is.na(INCOME)),
16     .groups = "drop"
17   )

```

I computed the number of congregations by religious classification in TRAD6 by year and then the mean and median income for the last fiscal year and dropped the NAs.

	YEAR	TRAD6	total_members
	<int>	<chr>	<dbl>
1	2009	Chr étiennes	0
2	2009	Juives	0
3	2009	Musulmanes	0
4	2022	Chr étiennes	1448291.
5	2022	Juives	35083
6	2022	Musulmanes	7433

	TRAD6	mean_income	median_income	count
	<chr>	<dbl>	<dbl>	<int>
1	Chr étiennes	474601.	201000	676
2	Juives	2332500	115000	4
3	Musulmanes	77941.	42500	17

4. Create a categorical variable for called `AVG_INCOME` that is binary in which 1 = "Above average or average income" and 0 = "Below average income", which indicates if a congregation is $\geq$ average income or $<$ average income among congregations that year.

```

1 Income_last_year
2 Religion_with_income_cat <- Religion %>%
3   group_by(YEAR) %>%
4   mutate(

```

```

5   avg_income_year = mean(INCOME, na.rm = TRUE) ,
6   AVG_INCOME = ifelse(INCOME >= avg_income_year, 1, 0)
7 ) %>%
8 ungroup()
9 Religion_with_income_cat$AVG_INCOME

```

I created code to see if the income of a congregation is average income or above average income based on other congregations.

Data Visualization

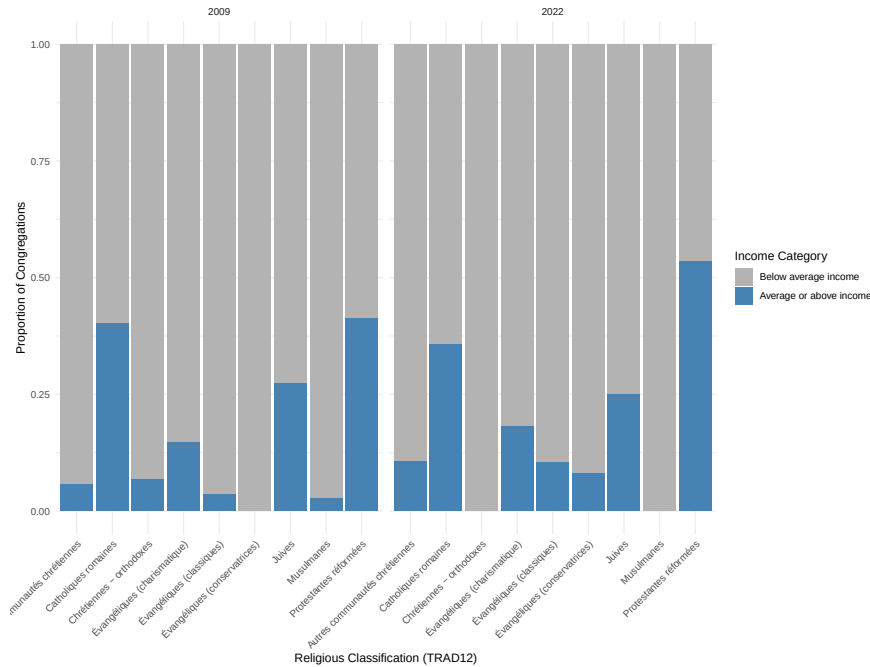
1. Create a bar plot visualizing the proportion of congregations above and below the average income (AVG_INCOME) in each year by 12-level religious classification (TRAD12). Hint: Use `facet()` for YEAR.

```

1 plot_data <- Religion_with_income_cat %>%
2   filter(!is.na(TRAD12), !is.na(AVG_INCOME)) %>%
3   group_by(YEAR, TRAD12, AVG_INCOME) %>%
4   summarise(n = n(), .groups = "drop") %>%
5   group_by(YEAR, TRAD12) %>%
6   mutate(prop = n / sum(n))
7 library(ggplot2)
8
9 income_plot <- ggplot(plot_data, aes(x = TRAD12, y = prop, fill = factor(
10   AVG_INCOME))) +
11   geom_col(position = "stack") +
12   facet_wrap(~ YEAR) +
13   labs(
14     x = "Religious Classification (TRAD12)",
15     y = "Proportion of Congregations",
16     fill = "Income Category"
17   ) +
18   scale_fill_manual(
19     values = c("0" = "gray70", "1" = "steelblue"),
20     labels = c("Below average income", "Average or above income")
21   ) +
22   theme_minimal() +
23   theme(
24     axis.text.x = element_text(angle = 45, hjust = 1)
25   )

```

I created a bar plot showing the proportion of congregations above and below the average income



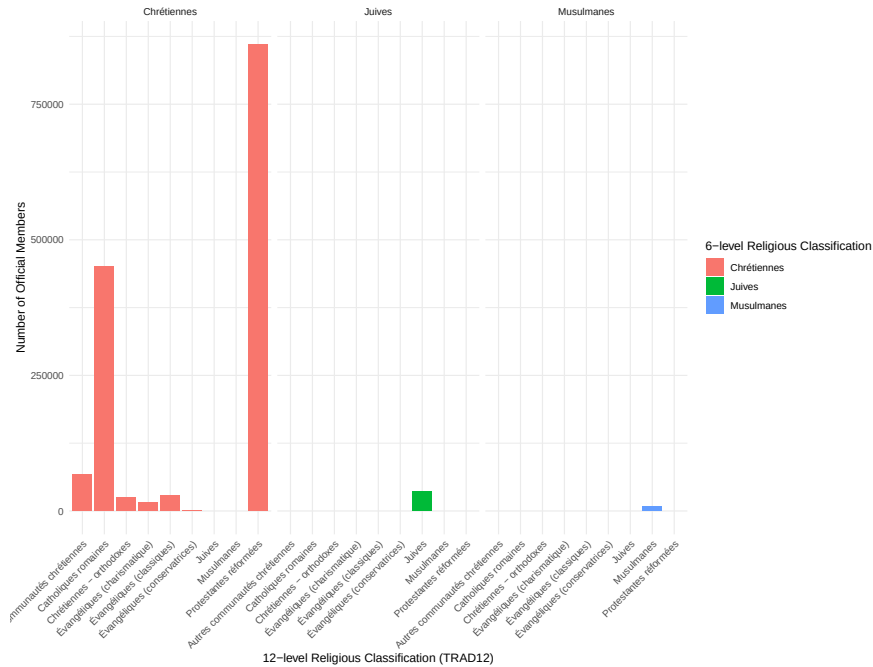
2. Make a histogram using `geom_col()` detailing the number of official members using the 12-level religious classification (TRAD12) distinguishing between the 6-level religious classification (TRAD6) in 2022. Hint: Use `facet()` for TRAD6, with TRAD12 on the x-axis in addition to group/fill with the `position="dodge"`.

```

1 hist_data <- Religion %>%
2   filter(YEAR == 2022,
3         !is.na(TRAD12),
4         !is.na(TRAD6)) %>%
5   group_by(TRAD12, TRAD6) %>%
6   summarise(
7     total_members = sum(NUMOFFMBR, na.rm = TRUE),
8     .groups = "drop"
9   )
10 religion_graph2 <- ggplot(hist_data, aes(x = TRAD12, y = total_members,
11     fill = TRAD6)) +
12   geom_col(position = "dodge") +
13   facet_wrap(~ TRAD6) +
14   labs(
15     x = "12-level Religious Classification (TRAD12)",
16     y = "Number of Official Members",
17     fill = "6-level Religious Classification"
18   ) +
19   theme_minimal() +
20   theme(
21     axis.text.x = element_text(angle = 45, hjust = 1)
22   )

```

I made a histogram for the number of official members by religious classification



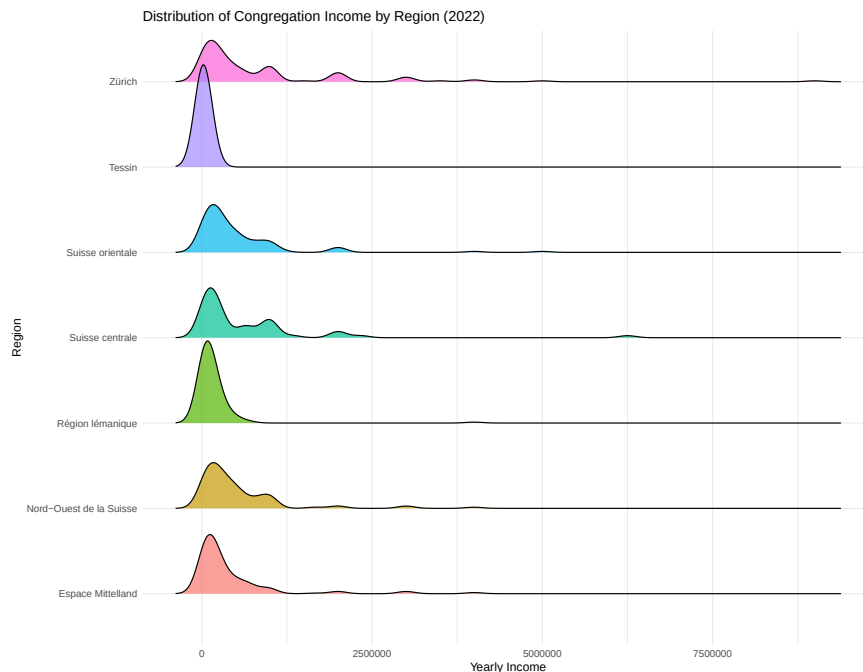
3. Display the distribution of yearly income (INCOME) in 2022 for congregations in each region (GDREGION) using ridge plots.

```

1 ridge_data <- Religion %>%
2   filter(
3     YEAR == 2022,
4     !is.na(INCOME),
5     !is.na(GDREGION)
6   )
7 library(ggribes)
8
9 religion_graph3 <- ggplot(ridge_data, aes(x = INCOME, y = GDREGION, fill =
10   GDREGION)) +
11   geom_density_ridges(alpha = 0.7, scale = 1.2) +
12   labs(
13     x = "Yearly Income",
14     y = "Region",
15     title = "Distribution of Congregation Income by Region (2022)"
16   ) +
17   theme_minimal() +
18   theme(
19     legend.position = "none"
20   )

```

I created a ridge plot to display the distribution of yearly income in 2022 for congregations in each region.



4. Create a boxplot of the number of official members per congregation in 2022 by religious classification (TRAD6) and region (GDREGION). Hint: Use `facet()` for GDREGION.

```

1 boxplot_data <- Religion %>%
2   filter(
3     YEAR == 2022,
4     !is.na(NUMOFFMBR),
5     !is.na(TRAD6),
6     !is.na(GDREGION)
7   )
8 religion_graph4 <- ggplot(boxplot_data, aes(x = TRAD6, y = NUMOFFMBR, fill
9   = TRAD6)) +
10  geom_boxplot() +
11  facet_wrap(~ GDREGION) +
12  labs(
13    x = "6-level Religious Classification (TRAD6)",
14    y = "Number of Official Members per Congregation",
15    title = "Distribution of Congregation Size by Religion and Region",
16    fill = "TRAD6"
17  ) +
18  theme_minimal() +
19  theme(
20    axis.text.x = element_text(angle = 45, hjust = 1),
21    legend.position = "none"
22  )

```

I created a boxplot for the members per congregation in 2022 by religious classification

